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SEAT No. :

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**PF199**

**APR-26/SE/Insem-245**

**S.E. (Civil Engg.) (Insem)**

**PROJECT MANAGEMENT**

**(2019 Pattern) (Semester - IV) (201012)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if required.

**Q1)** a) What are the scientific methods of project management? Explain with the help of example. [5]

b) Write a short note on Categories of project. [5]

c) Explain the following principles of management [5]

i) Unity of direction

ii) Esprit de Corps

OR

**Q2)** a) Explain with a neat sketch “Matrix Organization Structure” [5]

b) What do you mean by project life cycle? What are the different phases considered? [5]

c) What is meant by authority? Also state meaning of delegation of authority. [5]

**P.T.O.**

**Q3) a)** Listed below are the activities of a project:

**[8]**

| Activity | Duration (days) |
|----------|-----------------|
| 1-2      | 8               |
| 2-3      | 12              |
| 2-4      | 12              |
| 3-5      | 0               |
| 4-5      | 0               |
| 3-6      | 16              |
| 5-7      | 20              |
| 4-8      | 12              |
| 6-9      | 16              |
| 7-9      | 10              |
| 8-9      | 16              |

i) Draw a network, calculate total project duration and highlight critical path.

ii) Calculate EST, EFT, LST, LFT and total float for the activities.

b) Define

**[4]**

i) Optimistic time

ii) Most likely time

iii) Pessimistic time

iv) Expected mean time

c) Compare CPM and PERT

**[3]**

OR

- Q4) a)** The following table gives the time estimates for various activities of project: [8]

| Activity | Duration in weeks |       |       |
|----------|-------------------|-------|-------|
|          | $t_o$             | $t_m$ | $t_p$ |
| 1-2      | 1                 | 2     | 3     |
| 2-3      | 3                 | 6     | 9     |
| 2-4      | 2                 | 4     | 6     |
| 3-5      | 4                 | 6     | 8     |
| 4-6      | 4                 | 6     | 8     |
| 5-6      | 0                 | 0     | 0     |
| 5-7      | 3                 | 4     | 5     |
| 6-7      | 2                 | 5     | 8     |

- i) Draw the project network and find total project duration and highlight critical path.
  - ii) Calculate variance of all the activities.
  - iii) What is the probability that project will be completed in 19 weeks.
- b) What are the rules for drawing network diagram? [4]
- c) Define [3]
- i) Predecessor activity
  - ii) Successor activity
  - iii) Concurrent activities.

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